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An Automated Student Registration System: A Case Study of Lugazi Primary School

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ABSTRACT

The study focused on the developing an automated student registration System for the for Lugazi primary school. An automated student registration System is the one that provides for management oriented reporting, for example a DBMS. A DBMS is a software program designed to assist in maintaining and utilizing large amount of data for the purpose of management in an organization, The DBMS enabled maintaining, organizing and retrieving information efficiently, and these reports are usually generated from a shared database that stores data from many sources. The aim of this study was to analyze and design an automated student registration application which is a DBMS that addresses the problems of delay in student registration process, security of students' results, searching of information, efficiency in report generation for the student and also the difficulties faced while updating student's record in the school. With the obvious advantages that have accrued to the school in terms of faster time response, reduced volume of paper work, improved decision making, improved data integrity and security, resulting into saving costs in stationery usage, it is highly recommended that further developments on the system is needed to Changing the back end to SQL and Wamp Server, Adjusting the system to new requirements or re-designing or modifying the application system in response to constantly changing environment. It is evident that it will result in savings, which will lead to improved data storage, and also improved performance of the school.

Keywords: automated student registration System, and DBMS

INTRODUCTION

Information Technology is an important aspect of the business world today since organization must keep up with the competitive market hence Information Technology helps to achieve this through security, confidentiality, integrity of data, fast retrieval of information and manipulation of data, and improved customer satisfaction among the many benefits [1,2,3,4,5].

Lugazi primary school is a government aided day school, which started in 1989. The school is located in Mukono District. It is headed by Mrs. Mawanda Irene, deputized by Mrs. Mwaka Angela and Mr. Ssenyonga Andrew. It has 21 staff members of which 16 are teaching staff (11 females and 5 males) and the rest non-teaching-staff. Lugazi primary school has a population of 685 pupils with 308 boys and 377 girls. The school runs from P.1 to P.7. subjects taught by the school includes: English, Mathematics, Science, Social Studies, Integrated performance Science, Physical art and physical education (PAPE), Literacy Luganda reading and writing mainly for lower primary that is P.1 to P.3. Lugazi primary school has drawn in variety of parties that is Parents and charity organizations such as Inter-aid caters for refugees from Congo and Sudan. TASO caters for AIDS victimized pupils. Pupils pay school fees Ugandan shilling 25,000, in addition to UPE support from the government.

The school participates in all activities especially extra-curriculum activities such as Athletics, Music and sports, and has an outstanding record in music in the past three years in the division. General

academic performance is on average, for instance in the external examinations- Primary Leaving Examination candidates pass with 55% in division 1.

The school management system is manual (no computer is involve in storage and manipulation of pupil's information).

Records kept follow this category:

I. Registers per class for taking pupils attendance each day
II. Fees register book
III. Enrolment book
IV. Newcomers file
V. Children with disabilities file
VI. Personal teachers file

The manual system was well arranged/organized but faced some shortcomings.

I established the problem and found out workable solutions.

The school management finds it necessary to adapt to a new technological system. In relation to my study a database is more preferable, because it can handle and manage the following:

I. Tracking personal record of pupils
II. Ease searching and updates of pupils records
III. Provide better security and control over information

Statement of the problem

Lugazi primary school use a manual filing system for management and storage of information on pupils, employees and financial accounts, as well as pupil's report card generation as a result report cards are not produced in time. The current method of handling results at Lugazi primary school has problems such as lack of data integrity and accuracy in producing report cards for the pupil. Since the whole process is manual, modification or updates of pupil's results are done manually which may not be accurate due to human errors during typing and calculation of pupil's marks during results processing.

Aim of the study

To design an automated student registration system in lugazi primary school.

Specific objective

To develop a model database management system.

METHODOLOGY

Systems Development Life Cycle (SDLC) or sometimes just (**SLC**) is defined by the U.S. Department of Justice (DoJ) as a software development process [2]. According to https://i7lp.integral7.com/durango/do/login?ownername=cisco&channel=cisco&baseechannel=integral7 (Last viewed on 2nd June 2008: 10:35 am), the cycle is the process by which the system is conceived, developed and implemented [3]. The concept formation includes the statement of the findings, analysis of the findings, stating of user and system requirement and specifications. In the following chapter, system design, implementation, testing and system implementation will follow.

Findings from respondents

The problems that arise in the current management system of Lugazi primary school can be rectified to hold back the problems of storage, retrieval and security of information stored manually on papers.

Through my findings, I discovered some major expectations /needs of management that the current system does not offer efficiently to facilitate management of the school with regard to pupils. It was therefore recommended through interviewing the head teacher, secretaries and class teachers and giving them questionnaires to introduce a computerized Database system to handle the operations in the report processing department instead of the manual system been used.

Below are the findings with respect to the response to the interviews and the questionnaires used, the researcher categorized the findings (data collected through interviewing and giving questionnaires), the secretary was interviewed on Tuesday 15th April 2008 from 10:00am to midday in her office;

I. Registration of pupil

Registration of pupils is done by the secretary; she explains that

registration is done at the beginning of each term for the continuing pupils; this information is kept in a registration file for each class. The problems they face is when there are errors and need to be corrected, it would require changing the whole record or using a whitewash which is costly.

The other registration is done for incoming pupils, who are normally given a file of their own for tracking purposes, at the same time they are allocated a class for which they qualified for and they are then registered with the particular class.

All the registration is done manually (the secretary has to write all the information needed while registering a pupil) and it involves a lot of paper work since the school does not have a computer system.

II. Managing files

According to the head teacher of Lugazi primary school interviewed on Monday 14th April 2008 at 10:00am, in his office at Lugazi primary school, files are kept in shelves physically stored in a cabinet. The problems they face in this method of storage is that, the files get old and needs upgrading (buying new files to replace the old one) which is costly for the school. The shelves can become damaged due to insects such as termites and cockroaches. Consequently if upgrade of the shelves is not done, it can lead to lost of data since it is the only place where all school files are kept. Hence it calls for a new and better method of storage to reduce the risk and expenses they enquire in the current method of file storage at the school.

III. Maintaining data integrity

With every process being performed manually, data integrity has been such a difficult one to maintain as explained by the school secretary. she said that most of the complains they receive are concerning errors on the report cards, mostly misspelling of names, miscalculation of pupils' marks and incorrect placement of pupils' marks on the report card, these are human errors which can be eliminated if a computerized system was used. She therefore recommends for a better system to improve and possibly remove the problems she is facing currently.

IV. Transaction management

The secretary interviewed in her office at Lugazi primary school on Tuesday 15th April 2008 at 10:00 am, narrates that if a parent comes for his/her child's report card, she has to go physically to the shelve and look for a file of the particular class, then she checks if the pupil did exams, if yes then she goes ahead to check if the report card was ready then hands it to the parent. If the report card was not ready then she gets marks for all subjects done by the student from the class teacher concern then compiles the marks after which she fills a ready printed report form, then takes it to be signed by the head teacher, after which it is given to the parent.

This process also describes what she goes through when the head teacher asks for a report on a class or pupil. She described the activities as been time consuming and not efficient. Hence recommend for a better approach to solve the problem.

V. Security

The class teacher for primary five (P5) interviewed on Tuesday 15th April 2008 at 11:30 AM in the staff room of Lugazi primary school, explains how they prevent their information from intruders. "The school has shelves in the head teacher's office and it is mostly accessed by the secretary who has the key to the room. But this does not stop other teachers who may not have access rights from accessing those files especially when the head teacher goes out for some time. This method of securing records does not put into consideration things like cockroaches and other insects that

can spoil the files making them unreadable".

System requirements

Information about the hardware and software required to run a software item. According to http://www.itsmarc.com/crs/manl1744.htm (Last viewed on July 3, 2008 11:40 am) [4], system requirements cover the functional requirements, non-functional requirements, Hardware specifications and software specifications.

Functional requirements for the system

In software engineering, a **functional requirement** defines a function of a software system or its component [5]. A function is described as a set of inputs, the behavior, and outputs. According to http://en.wikipedia.org/wiki/Functional_requirements (Last viewed on May 29, 2008: 6:20pm)

A functional requirement is a description of activities and services a system must provide [6]. These are normally identified in terms of inputs, outputs, processes and stored data that are needed to satisfy the system [7,8,9,10,11,12].

Below are the functional requirements of the system;

I. It should allow the entry of new student's information and adding new users into the systems database by a GUI.
II. The system should allow the entry of student's marks and calculates the grade obtained by the particular student.
III. The system should be able to generate reports on student's performances, based on school terms and subjects offered [8].

Non – functional requirements for the system

In systems engineering and requirements engineering, **non-functional requirements** are requirements which specify criteria that can be used to judge the operation of a system, rather than specific behaviors [9,13,14,15,16]. Accordingtohttp://en.wikipedia.org/wiki/Non-functional_requirement (Last viewed on 3 July 2008 16:00).Below are the proposed non-functional requirements for the system.

Reliability: The system should deliver the required functionality and performance to the user [10,17,18]. That's, the system should produce the desired results as expected by the client/user, it should be tolerant and not crash when left on or in use for long time [19].

Efficiency: The system should not make wasteful use of system resources like processing time and memory space. That's, the system should not take very long time when performing a task and it should work well event if other programmes are running at the same time. It should not take a lot of memory and processing time [20].

Portability: It should run on different environments for example hardware and software environments. The system should run effectively under minimum hardware requirements. It should as well work with most of the different versions of windows operating system effectively [21].

Usability: The system should be usable by the users intended for without difficulties effort. The system should be easy to use to simplify work for the client/users, hence making it useful and meeting the user's need [22].

Enhancement: The system should accept modification to provide new functions or changing existing functions. It should be possible to add new functionalities to the system and change some of its parts incase if there is need to do so depending on the user's requirements [23].

Hardware requirements for the system

The following are the minimum hardware needed for the developed system;

I. One computer with basic input device such as keyboard, mouse and all other necessary input devices (such as a printer and scanner) are required.
II. 128 MB of Memory recommended. Additional memory may be required, depending on Operating system requirements
III. Intel Pentium 800 MHz or High microprocessor is recommended

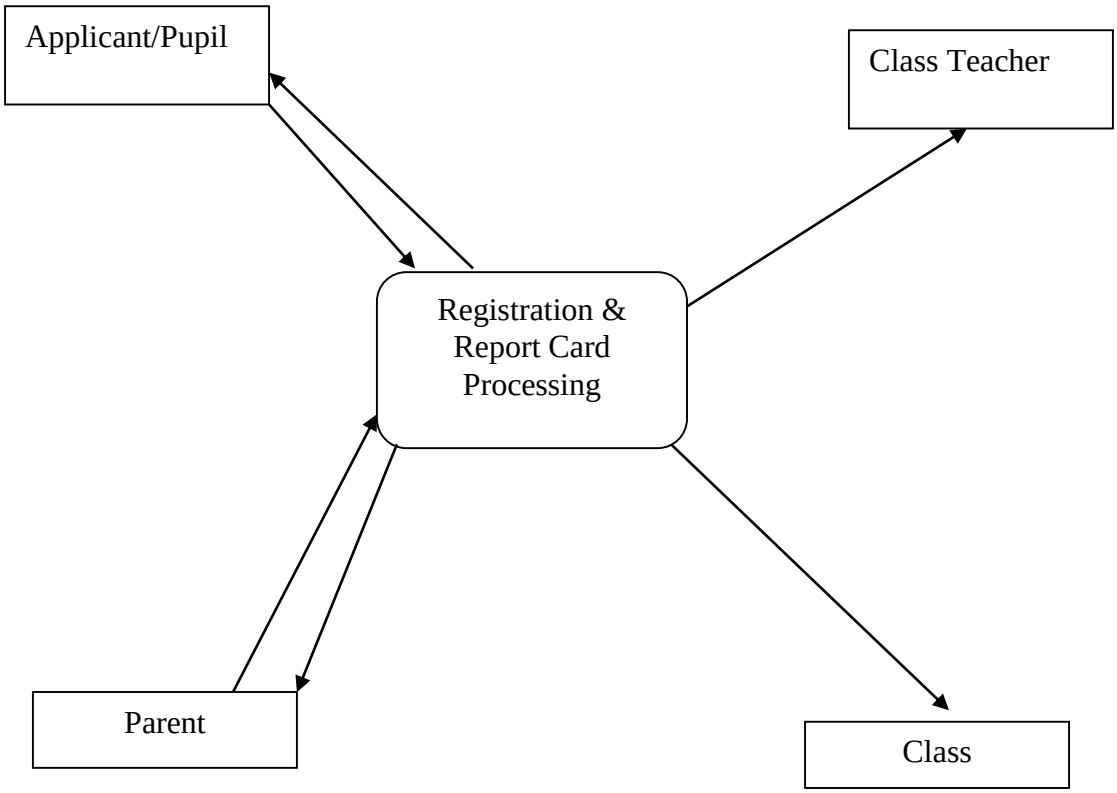
IV. VGA with resolution of 800x600 monitor or high is recommended and since the software requires printing reports and documents, a printer is needed [24].

Software requirements

The software will run in windows supporting environment. That means the operating system should be any of these;

Windows 9x (95 & 98)
Windows 2000
Windows XP

Figure 1: CONTEXT LEVEL DIAGRAM



Analysis of the manual system

The analysis of the manual system was made and categorized in weaknesses and strengths as explained below:

Strength of the manual system

Unlike a system crash which can be disastrous in electronic systems, the manual system here reduced the problem of complete data loss as there were several copies of data.

The electronic system environment is ever changing and this means its users have to update themselves else they are considered incompetent. However the manual system was not subjected to such changes.

Weakness in the manual system

Redundancy of data was very rampant in the manual system since information was duplicated in several places and manipulated by different person.

It was a tedious work for the staff to update information in the various locations in which they were stored at once. This further raised the chances of errors.

It was difficult to have data integrity thereby causing inaccurate information as a result. Security measures were also difficult to enforce since anybody could easily have access to any information they want.

Keeping of books and files occupied a lot of office space and without some kind of order; it took a long time to locate a particular piece of information as described by the school secretary.

The report generation department faced difficulties in processing report cards in time for the whole school and the parents have always complained of delay and inaccuracy in the report cards produced at the end of each term [24,25,26].

RESULTS

Logical design

The depiction of the conceptual design on a certain data structure (eg relational, object-oriented, network, transfer format). According to www.kadaster.nl/top10nl_engels/begrippenlijst.html (Last viewed on 3 July 2008, 17:36)

Logical database design identifies the important objects that need to be represented in the database and the relationships between these objects. It reviews the system requirements and considers the major components of the system.

Entities (Tables)

These are objects that can be distinguished from another; they cover the scope of the system design and are necessary for the overall management of the system. The DBMS has six entities, they include the following:

1 Student

This is where the student's information will be stored; it has twelve (12) attributes.

2 Department

This stores information of various departments in the school.

3 Staff

This will stores information about all employees of the school

4 Marks

This stores marks of the pupil scored in each subject offered in each term.

5 Guardian

This will store information about a guardian of each student

6 Security

This will store usernames and passwords

Physical Design

Describes how an information system will be physically implemented in order to meet its logical requirements. According to www.mckinnonsc.vic.edu.au/vceit/glossary.htm (Last viewed on 3 July 2008, 17:40)

Physical design describes how the logical design is to be physically implemented (as tables) in the target DBMS [24].

Conceptual model of the system

A conceptual model represents system data by entity and relationship between the entities. An entity was considered an object of interest in the design of the application while relationship is the interconnection of two entities. Conceptual modeling facilitates the design by allowing specification of the system schema that represents the overall structure of the system. During the application design, entity relation modeling was used and the following diagram was developed [25].

Figure 2: conceptual model

CONCEPTUAL MODEL, SHOWING RELATIONSHIP BETWEEN THE ENTITIES

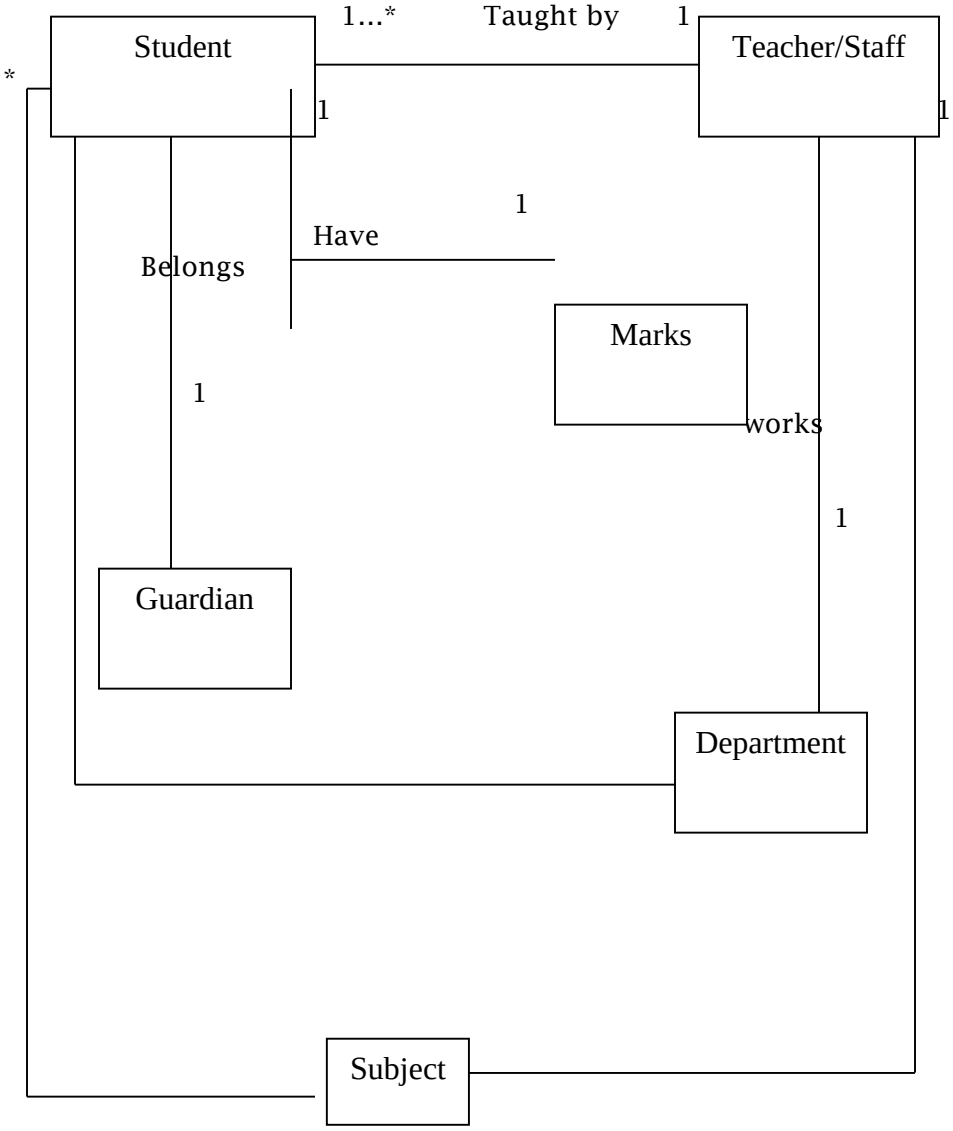


Table 1: Student Table showing the entities and the appended attributes.

Field name	*Data Type*	Size	*Constraints*	*Required*	*Description*
StudentID	Text	10	Primary Key	Yes	Student's identity number
FirstName	Text	20	Not Null	Yes	Student's First Name
OtherName	Text	20	Not Null	Yes	Student's Other Name
Sex	Text	10	Not Null	Yes	Student's Gender
Age	Number	10	Not Null	Yes	Student's Age
DateOfAdmission	Date/Time		Not Null	Yes	Date on which the student was Admitted
DateOfBirth	Date/Time		Not Null	Yes	Student's Date Of Birth
Country	Text	20	Not Null	Yes	Student's Country of Origin
Address	Text	20	Not Null	Yes	Student's postal Address
PhoneNumber	Number	15	Not Null	Yes	Student's Phone Number
Class	Text	15	Not Null	Yes	Student's Current Class
Email	Text	20	Null	No	Student's Email Address

Table 2: Department Table

Field name	*Data Type*	Size	*Constraints*	*Required*	*Description*
DepartmentID	Text	10	Primary key	Yes	Department Identity number
StaffID	Text	10	Foreign Key	Yes	Staff Identity Number
Departmentname	Text	20	Not Null	Yes	Department's Name
SubjectCode	Text	10	Foreign Key	Yes	Codes for the subject in the particular department

Table 3: Staff Table

Field name	*Data Type*	Size	*Constraints*	*Required*	*Description*
StaffID	Text	10	Primary Key	Yes	Staff Identity number
FirstName	Text	20	Not Null	Yes	Staff's First Name
OtherName	Text	20	Not Null	Yes	Staff's Other Name
Title	Text	10	Not Null	Yes	Staff Position or Title
Gender	Text	10	Not Null	Yes	Staff's Gender
Age	Number		Not Null	Yes	Staff's Age
Qualification	Text	20	Not Null	Yes	Staff's Qualification
DateHired	Date/Time		Not Null	Yes	Date on which the staff was Hired
Salary	Currency	10	Not Null	Yes	Staff's Salary
Phonenumber	Number	15	Not Null	Yes	Staff's Phone number
Address	Text	20	Not Null	Yes	Staff's Postal Address
Email	Text	20	Null	Yes	Staff's Email Address

Table 4: Marks Table

Field name	Data Type	*Size*	Constraints	Required	Description
Student ID	Text	10	Primary Key	Yes	Student's Identity Number
Term ID	Text	20	Foreign Key	Yes	Term Identification number
English	Number	5	Not Null	Yes	Marks for English
Math	Number	5	Not Null	Yes	Marks for Math
S.S.T.	Number	5	Not Null	Yes	Marks for S.S.T
Science	Number	5	Not Null	Yes	Marks for Science
Agriculture	Number	5	Not Null	Yes	Marks for Agriculture

Table 5: Guardian Table

Field name	Data Type	*Size*	Constraints	Required	Description
GuardianID	Text	10	Primary Key	Yes	Guardian's Identification number
FirstName	Text	20	Not Null	Yes	Guardian's First Name
OtherName	Text	50	Not Null	Yes	Guardian's Other Name
Sex	Text	10	Not Null	Yes	Guardian's Gender
Address	Text	50	Not Null	Yes	Guardian's Postal Address
StudentID	Text	10	Foreign Key	Yes	Student's Identity number
PhoneNumber	Text	15	Not Null	Yes	Guardian's Phone number
Email	Text	20	Not Null	No	Guardian's Email address

Table 6: Security

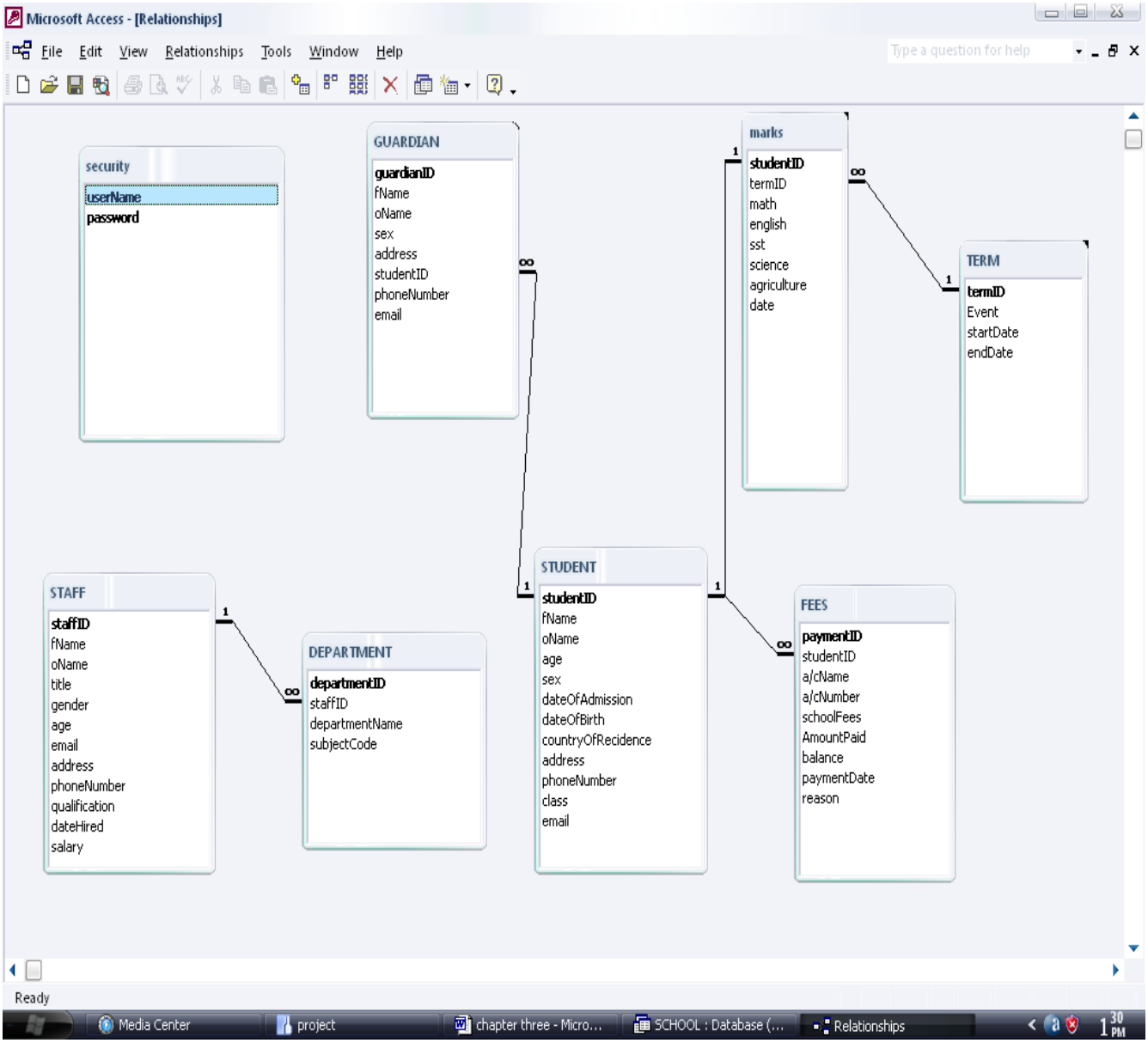
Field name	Data type	size	Constraints	Required	description
Username	Text	15	Primary key	yes	User name
Password	Text	15	Primary key	yes	User's password

Entity relationship model

An entity-relationship model (ERM) is an abstract conceptual representation of structured data [11]. Therefore an entity relations model represents system data by entity and relationship between the entities. An entity was considered an object of interest in the design of the application while relationship is the interconnection of two entities. Entity relationship modeling facilitates the design by allowing specification of the system schema that represents the overall structure of the system. During the application design, entity relation modeling was used and the following diagram was developed.

Figure 3: ERD

ENTITY RELATIONSHI DIAGRAM, SHOWING RELATIONSHIP BETWEEN THE ENTITIES



Interfaces

An interface is the aggregate of means by which people (the users) interact with a particular machine, device, computer program or other complex tool (the system). The user interface provides means of:

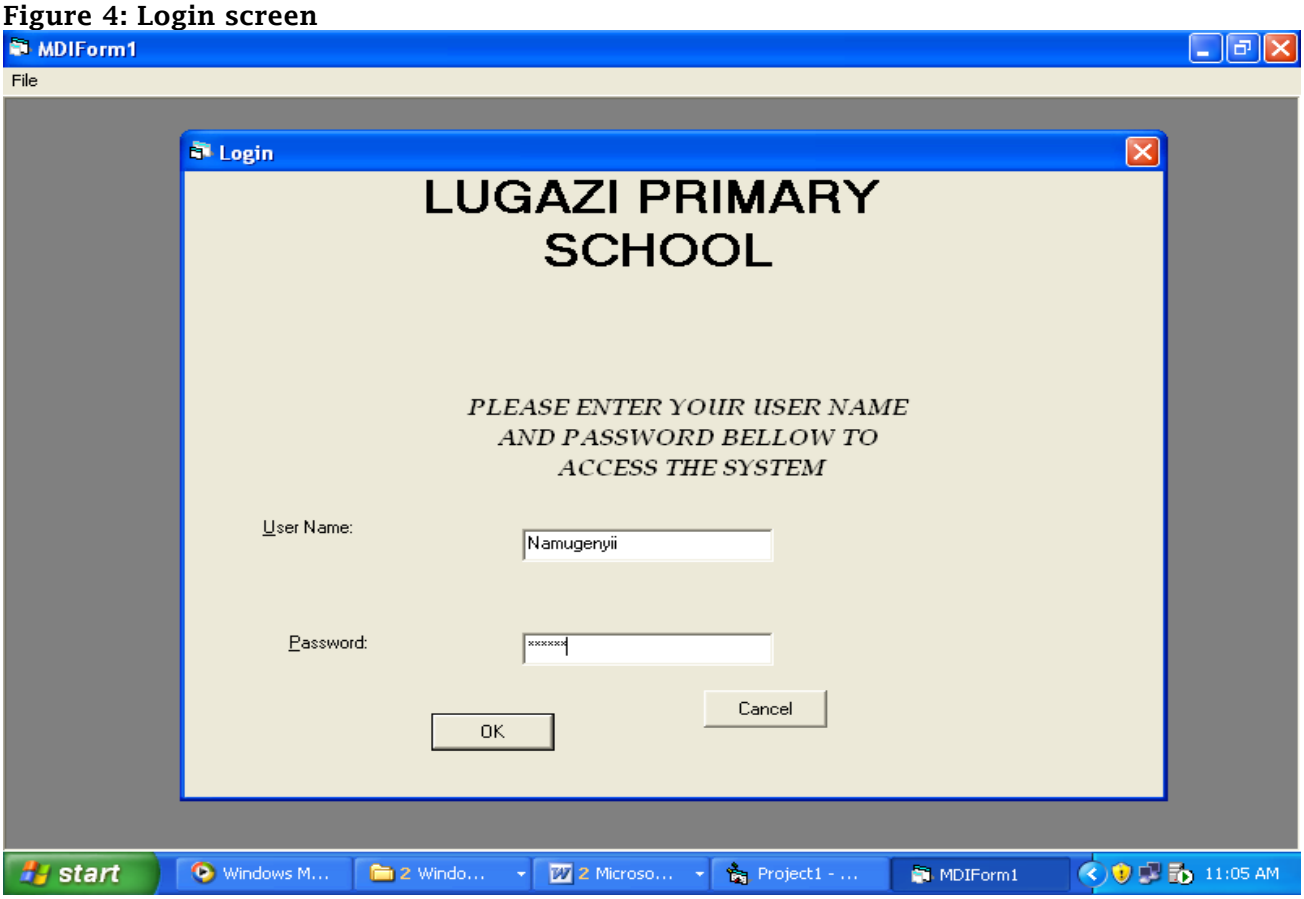
Input, allowing the users to control the system

Output, allowing the system to inform the users (also referred to as feedback). Accordingtohttp://bokardo.com/archives /definition-interfaces/(Last viewed on May 29, 2008: 6:10pm)

An interface is specified by a set of objects, the properties and the methods that they expose. The record set object will contain methods such as update, delete, save, search and exit. This is basically what the

users will interact with. Data will be captured and contained in one main control called forms. These interfaces will provide a neat presentation of information on initiative data entry process, precise way of capturing data in an environment that is user friendly while accessing and manipulating the stored in the database.

Figure 4: Login screen



Coding

Coding is the process of translating from a detailed design to statements in a programming language. According to http://www.google.co.ug/search?hl=en&defl=en&q=define:Coding&sa=X&oi=glossary_definition&ct=title(Last viewed on Thursday May 29, 2008: 6:20 pm).

This is a stage where the defined procedures are transformed into control specifications by the help of the computer language. This is the programming phase in which the researcher converted the program specification into computer instructions, which we refer to as programs. It is the programs that coordinate the data movement and control the entire process in a system. The following are the codes used for the various command buttons:

This system was programmed using the Visual Basic 6.0 in developing the interfaces. Because it is user-friendly programming language and it is easy to train. The database was designed using MS Access.

Sample codes

CODES FOR THE LOGIN FORM

CANCEL BUTTON

```
Private Sub cmdCancel Click()
   'Set the global var to false
   'To denote a failed login
   LoginSucceeded = False
   Me.Hide
End Sub
```

OK BUTTON

```
Private Sub cmdOK_Click()
```

```
Adodc1.Recordset.MoveFirst
While Not Adodc1.Recordset.EOF
If LCase (userNameSeen.Text) = LCase(Adodc1.Recordset.Fields(0)) And passwordSeen.Text = Adodc1.Recordset.Fields(1) Then
SCHOOL.Show
Exit Sub
Else
Adodc1.Recordset.MoveNext
End If
Wend
MsgBox ("incorrect password or username")
End Sub
```

TESTING

Types of system test include;

Functional Test

Functional test can be defined as testing two or more modules together with the intent of finding defects, demonstrating that defects are not present, verifying that the module performs its intended functions as stated in the specification and establishing confidence that a program does what it is supposed to do.

Acceptance Testing

Testing the system with the intent of confirming readiness of the product and customer acceptance.

Ad Hoc Testing

Testing without a formal test plan or outside of a test plan. With some projects this type of testing is carried out as an adjunct to formal testing. If carried out by a skilled tester, it can often find problems that are not caught in regular testing. Sometimes, if testing occurs very late in the development cycle, this will be the only kind of testing that can be performed. Sometimes ad hoc testing is referred to as exploratory testing [13].

Unit Testing

Unit testing is a test (often automated) that validates those individual units of source code is working properly. A unit is the smallest testable part of an application [14].

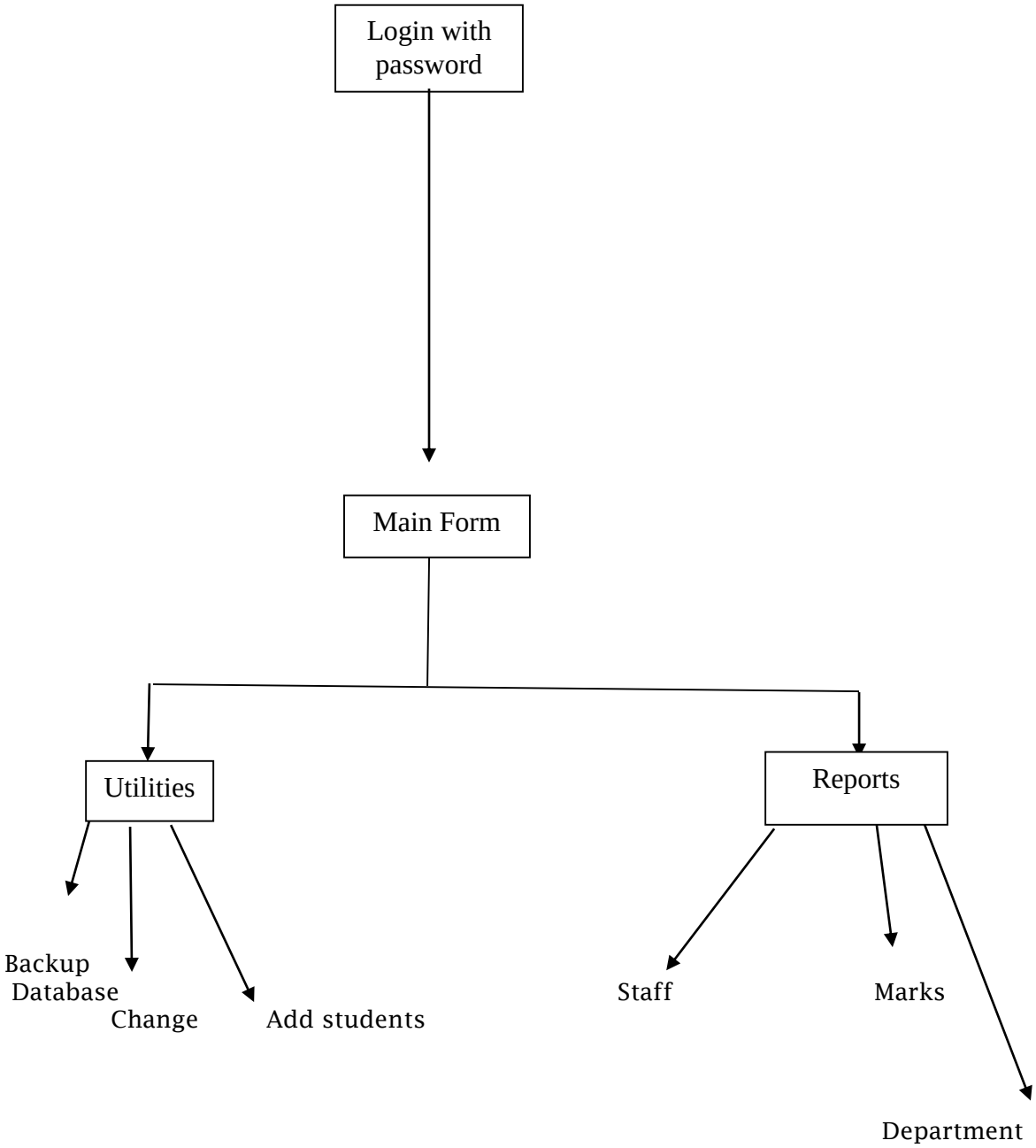
This was the first stage of testing; it was using written test plan and prepared test data. All the forms that are on the system were tested against the test plan and the conditions. The database was also tested and all the validation rules checked to see their effectiveness and efficiency.

System Testing

System testing of software or hardware is testing conducted on a complete, integrated system to evaluate the system's compliance with its specified requirements. System testing falls within the scope of black box testing, and as such, should require no knowledge of the inner design of the code or logic [15]. According tohttp://en.wikipedia.org/wiki/System_testing (Last viewed on May 27, 2008: 11:00 AM).The parts of the new program were link together and the school data was used to test the system. The computer system was tested to make sure it can function effectively and it can be able to come up with the expected output. Also massive amount of data was used to see if the system could be made to fail [25,26,27,28,29].

Working of the developed system

Figure 5: working model of the developed system
Below represents the different activities that can be performed by the system



System Implementation

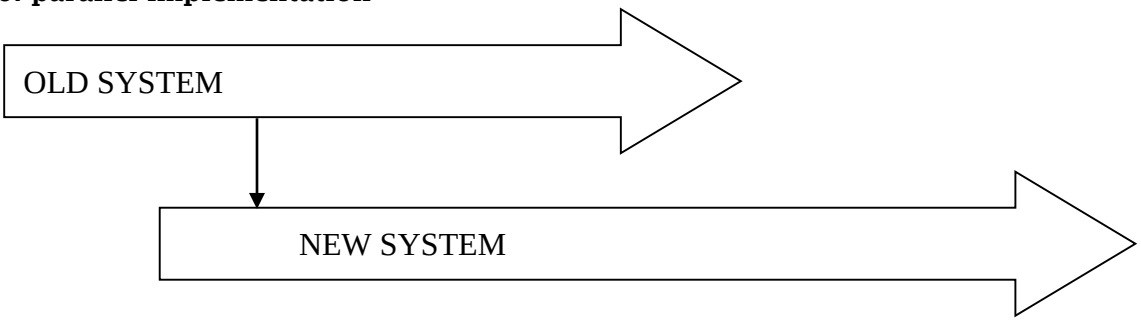
System Implementation involves putting the planned system into action. The stage of systems development in which hardware and software are acquired, developed and installed, the system is

tested and documented, people are trained to operate and use the system, and an organization converts to the use of a newly developed system [16]. According to http://www.bitpipe.com/tlist/Systems-Implementation.html (Last viewed on May 27, 2008: 11:10 AM)

Parallel Implementation

This is a technique where the old system phased out and the new system is introduced and operated side by side until the new system is proved reliable. The researcher used the parallel implementation technique to put the new system in place. Here the new system was implemented side-by-side with the old system. The benefit in taking this cautious approach is that if the new system fails the school can switch back to the old system.

Figure 6: parallel implementation



Hardware conversion

Computer system conversion is changing the computer model and peripheral devices [17]. Since there were no computers at the school, a new one was bought with minimum requirements to run the application. It was then tested by the researcher to ensure that all hardware and software components are functioning as expected.

Software conversion

Software conversion is the process of changing application programs to link to a new operating system or DBMS [18].

The DBMS program was installed and configured on the computer system. It was tested by the researcher to make sure the program works well on the new system and an operating system that meets the applications' minimum requirements was stalled.

System Maintenance

The use of the word maintenance to describe activities undertaken on software systems after delivery has been considered a misnomer due to its failure to capture the evolutionary tendency of software products [12].

System maintenance was done through the following;

Audit

Is the independent review and examination of records and activities to assess the adequacy of system controls, to ensure compliance with established policies and operational procedures, and to recommend necessary changes in controls, policies, or procedures. According to www.pki.vt.edu/help/glossary.html (Last viewed on May 27, 2008: 11:30 AM).

This is where independent review of the database management system to see whether all the records and the system are as they should be. The researcher designed an audit trail to help independent auditors to trace and record an action from the DBMS output through all the processing and storage to the source.

System Control

Forms and command buttons will be used to control the DBMS. Data will be input through the forms design by researcher and will be stored in the database tables.

Command buttons will be used to perform functions such as saving, canceling, deleting, adding new record and search for a record in the database tables.

System Backup

In information technology, **backup** refers to making copies of data so that these

additional copies may be used to restore the original after a data loss event. According to http://en.wikipedia.org/wiki/Backup (Last viewed on May 27, 2008: 11:30 AM)

Users shall protect their copy of the VB project file using their computer's file system and periodically make backup. In the event that the project file is accidentally altered, users can recopy the file from their backup copy. The backup utility on the computer system will be use to create a copy of the data on the hard disk. In the event that the original data on the hard disk is accidentally erased or over written or becomes inaccessible because of the hard disk malfunction, the copy will be used to restore the lost or damaged data.

Access Control

Access Control is any mechanism by which a system grants or revokes the right to access some data, or perform some action. Normally, a user must first Login to a system, using some Authentication system. Next, the Access Control mechanism controls what operations the user may or may not make by comparing the User ID to an Access Control database. According to http://hitachi-id.com/concepts/access_control.html (Last viewed on May 27, 2008: 12:00 AM). Access is restricted to only those who have a legitimate right to access the database management system. This helps prevent unauthorized users from accessing the data in the database management system. This was done through creating usernames and password to the authorized user of the application.

CONCLUSION

The design and implementation of the application was a very interesting venture that irrespective of some difficulties especially due to unavailability of enough resources(money to facilitate the researcher) and time to carry out adequate study on the case study, it was successful. As per the current design, the information about the students, parents/guardians and the staff of the school shall be confined within one computer system. Thus the staff of the school can only retrieve, update and delete any information regarding a student or staff through the secretary of the school.

After integrating all separate tested modules together, the system has satisfied the user requirements since the manual system appear no longer adequate and satisfactory for any organization that wants to accomplish its objectives by making its services quick, accurate, secure and up to date.

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